Ferdi Caner Bayram

Mechanical and Materials Engineer

Mechanical and Materials Engineer (M.Sc.) with hands-on experience in mechanical design, manufacturing, and industrial project execution. Skilled in preparing technical drawings, performing structural and thermal analyses, selecting performance-based materials, and ensuring compliance with ASME and industry standards. Proficient in SolidWorks, Inventor, ANSYS, and Abaqus for simulation-driven design validation. Experienced in collaborating with production and quality teams to solve complex engineering challenges and deliver reliable, cost-effective solutions, while leveraging AI-supported analytical tools to enhance problem-solving and process optimization.

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Fort Lee, NJ, United States

WORK EXPERIENCE

Mechanical Engineer

Adams International LLC

12/2024 – Present *Remote*

Independent distributor of electronic, electrical, and electromechanical components, offering global sourcing, inventory, and supply-chain solutions from over 1,000 trusted brands to help manufacturers and OEMs keep production running.

Achievements/Tasks

- Designed and improved mechanical components and assemblies using CAD tools, ensuring compliance with industry standards and manufacturing feasibility.
- Identified root causes of production and design issues through technical analysis and implemented practical solutions to enhance reliability and product performance.
- Collaborated with cross-functional teams (engineering, production, quality, and procurement) to support project execution from concept through fabrication.
- Prepared and reviewed engineering documentation including technical drawings, material specifications, calculations, and inspection reports to maintain quality and regulatory compliance.

Project Engineer

Piping Technology & Products Inc.

10/2022 – 12/2024 *Houston, TX, United States*

PT&P and its wholly-owned subsidiaries manufacture a wide range of high-quality engineered pipe supports.

Achievements/Tasks

- Designed expansion joints in compliance with EJMA, ASME BPVC Section VIII, and ASME B31.3, performing stress evaluations, design calculations, and material selections aligned with customer requirements and applicable codes and standards.
- Developed cost estimates and technical quotations by analyzing customer specifications, drawings, and project requirements, leveraging knowledge of materials, manufacturing processes, and labor hours to deliver accurate, competitive proposals.
- Generated over $3 million in purchase orders by partnering with the sales team to deliver technically compliant, cost-efficient proposals tailored to client needs.
- Collaborated directly with clients to interpret and clarify technical requirements, resolve design questions, and provide guidance on expansion joint selection and application.

- Coordinated cross-functionally with sales, drafting, production, procurement, and quality teams to ensure seamless project execution from proposal through delivery.
- Provided technical support during fabrication and reviewed designs to maintain on-time delivery and full compliance with project specifications.

Mechanical Engineer
Kumtel Home Appliances Co.

12/2021 – 05/2022 *Kayseri, Turkey*

Turkey-based white goods manufacturer, among Turkey's top 500 industrial enterprises since 2005, exporting to over 120 countries.

Achievements/Tasks

- Drove a 15% improvement in production line efficiency across high-volume home appliance manufacturing through strategic process redesign, while maintaining zero deviations from safety and quality standards.
- Managed the full lifecycle of engineering projects valued at over $10 million, spanning design, production, and quality assurance from initial concept to final delivery.
- Collaborated with R&D and manufacturing to re-engineer mechanical components, cutting material costs by 10% and extending product lifespan.
- Reduced equipment downtime by 20% by establishing structured root-cause analysis and preventive maintenance frameworks to address recurring production failures.
- Led cross-department coordination meetings between production, R&D, and quality teams, accelerating product launch timelines and improving communication flow.
- Stood up a new laser-cutting department from the ground up, from equipment procurement and facility layout to workforce training, delivering a fully operational unit integrated into existing production workflows.

Mechanical Engineer
ELC Aluminum Inc.

07/2021 – 11/2021 *Kayseri, Turkey*

Metal production and aluminum profile processing company with over 40 years of experience in the sector.

Achievements/Tasks

- Improved first-pass yield by 20% by leading cross-departmental technical reviews that identified root causes of recurring production defects and drove corrective actions.
- Reduced material waste by designing and optimizing aluminum extrusion and fabrication components through targeted design refinements and tighter process controls, while raising overall product precision.
- Accelerated production throughput by working with production and quality teams to streamline CNC machining and assembly workflows, cutting cycle times.
- Performed thermal and structural analyses on aluminum profiles and mechanical assemblies using SolidWorks, AutoCAD, and ANSYS, validating designs against strength and tolerance requirements.
- Contributed to over $2 million in customer-specific projects for architectural and industrial clients, supporting the full project lifecycle from initial design and prototyping through final delivery.

Researcher
Researcher Training Program for Defense Industry (SAYP)

07/2019 – 06/2021 *Kayseri, Turkey*
Abdullah Gul University · Supported by the Presidency of Defense Industries (SSB)

Project: Determination of hydrogen embrittlement behavior of armor steels by experimental methods and the optimization of hydrogen back-diffusion operation.

Achievements/Tasks

- Investigated the effect of hydrogen embrittlement on the mechanical properties of armor steels used by FNSS Defense through tensile, compression, Charpy V-notch, high strain rate, and ballistic tests.
- Optimized the time and temperature parameters for the back-diffusion of hydrogen atoms (bake-out process) introduced into the microstructure by electrochemical charging.

Researcher
The Scientific and Technological Research Council of Turkey (TUBITAK)

09/2018 – 08/2019 *Kayseri, Turkey*
Abdullah Gul University

Project: The effects of different microstructural variables on the hardening behavior of high manganese Fe-33Mn-Cx steels (Project No: 118M448).

Tasks

- Investigated the effect of different microstructural variables on the hardening behavior of high manganese Fe-33Mn steels using Visco-Plastic Self-Consistent (VPSC) algorithms.
- Carried out crystal plasticity modeling of iron-balanced austenitic steel with 33% manganese (Fe-33Mn) using experimental inputs.

SKILLS

Design & Simulation
SolidWorks | Autodesk Inventor | AutoCAD | ANSYS | Abaqus | MATLAB | FEA / Thermal & Structural Analysis | GD&T

Codes & Standards
ASME B31.3 | ASME BPVC Section VIII | EJMA | Piping Design

Materials
Material Selection & Metallurgy | Materials Characterization | Microstructural Evaluation | Mechanical Testing (Tensile, Compression, Ballistic)

Manufacturing & Quality
Production Line Optimization | Process Improvement | Root Cause Analysis & Corrective Action (RCCA) | Technical Documentation | Problem-Solving

Project & Commercial
Project Lifecycle Management | Project Leadership | Cross-Functional Team Leadership | Technical Proposals & Cost Estimation

Digital Tools
AI-Assisted Engineering Tools | Data-Driven Engineering | Microsoft Office (Excel, Word, PowerPoint)

EDUCATION

M.Sc. in Advanced Materials and Nanotechnology

Abdullah Gul University

09/2018 – 05/2021 *3.67 GPA, High Honor Degree · Kayseri, Turkey*

Thesis

- The Determination of Hydrogen Embrittlement Behavior of Armor Steels by Experimental Methods and the Optimization of Hydrogen Back-Diffusion Operation

B.Sc. in Mechanical Engineering

Abdullah Gul University

09/2013 – 06/2018 *3.21 GPA, Honor Degree · Kayseri, Turkey*

PUBLICATIONS

Journal · 2021

Investigations of strain rate, size, and crack length effects on the mechanical response of PCL electrospun membranes

Ferdi Caner Bayram, Mehmet Fazıl Kapçı, Adile Yuruk, Burak Bal, Ismail Alper Isoglu

Proceedings of the Institution of Mechanical Engineers, Part E: Journal of Process Mechanical Engineering · *IF 1.822*

Journal · 2020

Effect of hydrogen on fracture locus of Fe-16Mn-0.6C-2.15Al TWIP steel

Burak Bal, Barış Cetin, **Ferdi Caner Bayram**, Eren Billur

International Journal of Hydrogen Energy · *IF 7.2*

Journal · 2020

A detailed investigation of the effect of hydrogen on the mechanical response and microstructure of Al 7075 alloy under medium strain rate impact loading

Burak Bal, Bilge Okdem, **Ferdi Caner Bayram**, Murat Aydin

International Journal of Hydrogen Energy · *IF 7.2*

Journal · 2020

Lateral angular co-extrusion: geometrical and mechanical properties of compound profiles

Susanne Elisabeth Thürer, Julius Peddinghaus, Norman Heimes, **Ferdi Caner Bayram**, Burak Bal, Johanna Uhe, Bernd-Arno Behrens, Hans Jürgen Maier, Christian Klose

Metals · *IF 2.351*

M.Sc. Thesis · 2021

The Determination of Hydrogen Embrittlement Behavior of Armor Steels by Experimental Methods and the Optimization of Hydrogen Back-Diffusion Operation

Ferdi Caner Bayram

Abdullah Gul University